Annual Progress Monitor(APM) Report

QI CHEN

Feb 2026

1 Inference via Identity by Descent(IBD segment)

We will use the average fraction of the genome shared through Identity by Descent(IBD) segments of length between u centiMorgans and v centiMorgans. Here, u and v are given. The explicit formula for the expectation of the above statistics is as follows.

$$\mathbb{E}_{[u,v]}[f|\theta] = \frac{1}{|I|}\sum_{i\in I}\mathbb{E}[\mathbb{1}\{u \leq l_i \leq v\}|\theta] = \frac{1}{|I|}\sum_{i\in I}p(u \leq l_i \leq v|\theta) = p(u \leq L \leq v|\theta) \tag{1}$$

We treat the continuous genome as a discrete set I of sites. l_i is the length of the IBD segments containing site i. Thus, we are left to compute the probability that a site is contained in an IBD segment of length l, where l is between u and v.

If we are interested in the probability that IBD segments lie in the length range $R = [u, v]$ then

$$\begin{aligned}
p(L \in R|\theta) &= \int_{t_1=0}^{t_2=\infty}\int_u^v p(L = l|g)p(g|\theta)dldg \\
&= \int_{t_1=0}^{t_2=\infty} p(g|\theta)\int_u^v (\frac{g}{50})^2 l\exp(-\frac{g}{50}l)dldg \\
&= \int_{t_1=0}^{t_2=\infty} p(g|\theta)\left(\exp(-\frac{u}{50}g)(\frac{u}{50}g+1) - \exp(-\frac{v}{50}g)(\frac{v}{50}g+1)\right)dg
\end{aligned}$$

Thus, modeling the distribution of coalescence in different epochs will give us an explicit formula for the expectation of total fraction of the genome. We choose the Wright-Fisher model for the probability of coalescing, that is, time follows a exponential distribution with rate as the effective population size.

Suppose that we are interested in the contribution between time t_1 and t_2 where the effective population size is fixed to N, then we will have the explicit formula:

$$\begin{aligned}
g(t_1, t_2, N, u, v) &= \int_{t_1}^{t_2} p(t|\theta)\left(\exp(-\frac{u}{50}t)(\frac{u}{50}t+1) - \exp(-\frac{v}{50}t)(\frac{v}{50}t+1)\right)dt \\
&= s \times \int_{t_1}^{t_2}\frac{1}{N}\exp(-\frac{(t-t_1)}{N})\left(\exp(-\frac{u}{50}t)(\frac{u}{50}t+1) - \exp(-\frac{v}{50}t)(\frac{v}{50}t+1)\right) \\
&= s\exp(\frac{t_1}{N}) \times \left(\frac{u}{50N}\left(\left[\frac{\exp(c_u t)(c_u t-1)}{c_u^2}\right]_{t=t_1}^{t=t_2}\right) + \frac{1}{N}\left[\frac{\exp(c_u t)}{c_u}\right]_{t=t_1}^{t=t_2}\right) \\
&\quad - s\exp(\frac{t_1}{N}) \times \left(\frac{v}{50N} \times \left(\left[\frac{\exp(c_v t)(c_v t-1)}{c_v^2}\right]_{t=t_1}^{t=t_2}\right) - \frac{1}{N}\left[\frac{\exp(c_v t)}{c_v}\right]_{t=t_1}^{t=t_2}\right) \\
&= s \times f(t_1, t_2, N, u, v)
\end{aligned} \tag{2}$$

where $c_u = -\left(\frac{1}{N} + \frac{u}{50}\right)$, $c_v = -\left(\frac{1}{N} + \frac{v}{50}\right)$ and s represent a survival function that will be clear later. If we only consider one population, and we have the same population size throughout the time, then we can simply drop different epochs(since the term $\exp(\frac{t_1}{N})$ cancels out with s) and let t_1 be zero and t_2 be infinity. Working with a multi-population structure makes splitting time into different epochs inevitable.

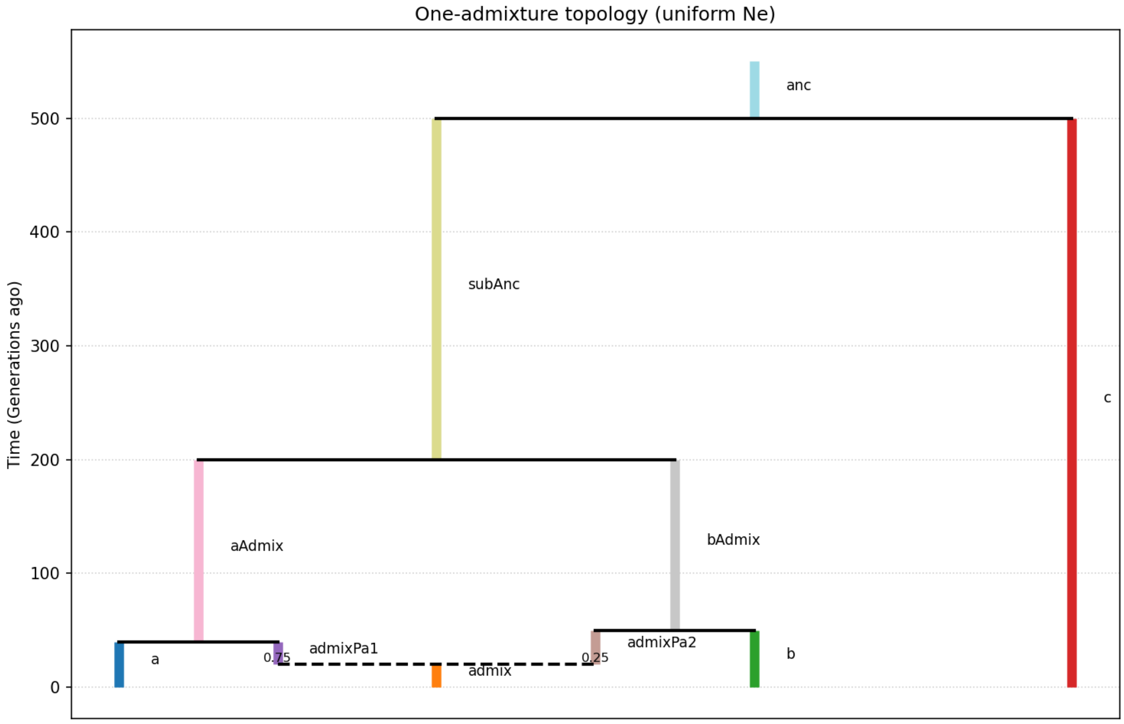

Figure 1: Example of population history

1 shows an example of the demographic structures we will work with. The population 'admix' splits into two populations, 'admixPa1' and 'admixPa2' backward in time, and they then merge with 'a' and 'b' separately, then merge to form 'subanc' and finally everything leads to 'anc'. We define each epoch as a continuous time interval separated by different events. For example, in 1 we have 6 epochs in total. We further assume that each event will generate a new population with its own population size, and population size is held constant for each population.

In order to model the probability distribution to the first coalescence time, we define the survival function

$$S(N, t_a, t_b) = \exp\left(-\frac{t_b - t_a}{N}\right)$$

as the total probability that two lineages in the same population of size N do not coalesce between time t_a and t_b.

As a working example, we derive the expectation of genome fraction shared through IBD segments between length u and v for population 'admix' and itself. Suppose that the admixture ratio is α to 'admixPa1' and thus $1-\alpha$ to 'admixPa2'. The first epoch will have contribution $f(0, t_1, N_{admix}, u, v)$, with survival probability $S(N_{admix}, 0, t_1)$ passed to the second epoch. The second epoch consists of the contribution from 'admixPa1' and the contribution from 'admixPa2'. The contribution is

$$S(N_{admix}, 0, t_1) \times \left(\alpha^2 f(t_1, t_2, N_{admixPa1}, u, v) + (1-\alpha)^2 f(t_1, t_2, N_{admixPa2}, u, v)\right)$$

For the third epoch we will have

$$\alpha^2 S(N_{admix}, 0, t_1) S(N_{admixPa1}, t_1, t_2) f(t_2, t_3, N_{aAdmix}, u, v)$$
$$+(1-\alpha)^2 S(N_{admix}, 0, t_1) S(N_{admixPa2}, t_1, t_2) f(t_2, t_3, N_{admixPa2}, u, v)$$

For the fourth epoch, this will be

$$\alpha^2 S(N_{admix}, 0, t_1) S(N_{admixPa1}, t_1, t_2) S(N_{aAdmix}, t_2, t_3) f(t_3, t_4, N_{aAdmix}, u, v)$$
$$+(1-\alpha)^2 S(N_{admix}, 0, t_1) S(N_{admixPa2}, t_1, t_2) S(N_{admixPa2}, t_2, t_3) f(t_3, t_4, N_{bAmix}, u, v)$$

Now for the fifth epoch, since there are three tracks for lineages from 'admix' to 'subAnc', we have three separate contributions:

$$\alpha^2 S(N_{admix}, 0, t_1) S(N_{admixPa1}, t_1, t_2) S(N_{aAdmix}, t_2, t_3) S(N_{aAdmix}, t_3, t_4) f(t_4, t_5, N_{subAnc}, u, v)$$
$$+(1-\alpha)^2 S(N_{admix}, 0, t_1) S(N_{admixPa2}, t_1, t_2) S(N_{admixPa2}, t_2, t_3) S(N_{bAdmix}, t_3, t_4) f(t_4, t_5, N_{subAnc}, u, v)$$
$$+2\alpha(1-\alpha) S(N_{admix}, 0, t_1) f(t_4, t_5, N_{subAnc}, u, v)$$

And finally, for the last epoch to ∞ time, we have

$$\alpha^2 S(N_{admix}, 0, t_1) S(N_{admixPa1}, t_1, t_2) S(N_{aAdmix}, t_2, t_3) S(N_{aAdmix}, t_3, t_4) S(N_{subAnc} t_4, t_5) f(t_5, \infty, N_{anc}, u, v)$$
$$+(1-\alpha)^2 S(N_{admix}, 0, t_1) S(N_{admixPa2}, t_1, t_2) S(N_{admixPa2}, t_2, t_3) S(N_{bAdmix}, t_3, t_4) S(N_{subAnc}, t_4, t_5) f(t_5, \infty, N_{anc},$$
$$+2\alpha(1-\alpha) S(N_{admix}, 0, t_1) S(N_{subAnc}, t_4, t_5) f(t_5, \infty, N_{anc}, u, v)$$

Taking the sum of the six epochs, we have the expectation we want. We could similarly calculate this statistic for any pair of populations given the topology.

1.1 Algorithm for Computing Expected Fraction

The worked example above makes clear that, written out by hand, the number of admixture/survival tracks grows combinatorially with the number of admixture events between the two sampled leaves and their most recent common ancestor. Rather than enumerating tracks, we compute the same sum by propagating a *joint lineage-location distribution* backward through the epochs with a linear-algebraic recursion. This is the form actually implemented in the Stan models and it applies to an arbitrary topology.

Topology as a sequence of transition matrices. Let the topology have n nodes (every leaf, every internal population created by an event, and the root). We order the events backward in time and associate to event e a transition matrix $M_e \in \mathbb{R}^{n\times n}$ acting on the node-occupancy of a single lineage:

- A **merge** event that sends children c_1, c_2 into parent p is the identity except that rows c_1, c_2 are redirected to column p: $(M_e)_{c_1,p} = (M_e)_{c_2,p} = 1$, with the original diagonal entries zeroed.
- An **admixture** (split) event that sends an admixed node c into parents p_1, p_2 with mixing fraction α is the identity except for row c: $(M_e)_{c,p_1} = \alpha$, $(M_e)_{c,p_2} = 1-\alpha$, with $(M_e)_{c,c} = 0$.

A single lineage with occupancy (row) vector $w^\top$ is updated across an event by $w^\top \leftarrow w^\top M_e$. Merge matrices are fixed by the topology; admixture matrices depend on the parameter α and are rebuilt every time the sampler proposes a new α.

Joint distribution of a pair of lineages. For a pair of sampled leaves (i, j) we track the matrix

$$W_{ab} \;=\; p\big(\text{lineage 1 in node } a,\ \text{lineage 2 in node } b,\ \text{no coalescence so far}\big),$$

initialized to the indicator $W = e_i e_j^\top$ (both lineages at their leaves). Because the two lineages move independently through the migration step, an event acts on W by the congruence

$$W \;\leftarrow\; M_e^\top \, W \, M_e.$$

The diagonal entry W_{aa} is exactly the probability that *both* lineages sit in node a at the start of an epoch (carrying all the admixture weights such as α^2, $(1-\alpha)^2$, $2\alpha(1-\alpha)$ that appeared by hand), having survived without coalescing in every earlier epoch. Only when both lineages are co-located can they coalesce, so only diagonal mass contributes to IBD.

Epoch accumulation and survival update. Within an epoch $[t_{\text{start}}, t_{\text{end}}]$ the function $f(t_{\text{start}}, t_{\text{end}}, N_a, u, v)$ of eq. (2) gives the *conditional* contribution to $\mathbb{E}_{[u,v]}[f]$ from a coalescence occurring in node a during that epoch, given both lineages entered a at t_{start}. The survival probability up to t_{start} is already carried inside W_{aa}, so the epoch contribution and the carried survival update are simply

$$\Delta\mathbb{E}_{[u,v]}[f] \;\mathrel{+}=\; \sum_a W_{aa}\, f(t_{\text{start}}, t_{\text{end}}, N_a, u, v), \qquad W_{aa} \;\leftarrow\; W_{aa}\, S(N_a, t_{\text{start}}, t_{\text{end}}),$$

performed once per epoch, after which the next event matrix is applied. The off-diagonal mass (lineages in different nodes) cannot coalesce and is therefore passed through unchanged by the survival step. The expected number-density quantity $\tilde{f}$ of eq. (3) is accumulated identically, giving the per-bin Poisson rates used in section 1.2. Running this recursion for every leaf pair (i, j) and every length bin $b = [u_k, v_k]$ produces the full set of expected fractions $\mathbb{E}_{[u_k,v_k]}[f_{i,j} \mid \theta]$.

Algorithm 1: Expected IBD fraction for all leaf pairs and length bins

Input: topology (ordered events $\{M_e\}$, node sizes $\{N_a\}$, epoch times $0 = \tau_0 < \tau_1 < \cdots < \tau_n < \tau_{n+1} = T_{\max}$), bins $\{[u_k, v_k]\}_{k=1}^K$
Output: $\mathbb{E}_{[u_k,v_k]}[f_{i,j}]$ for all leaf pairs (i, j) and bins k

Build parameter matrices M_e from the current $\{\alpha\}$ (admixture) and topology (merges)
foreach *leaf pair* (i, j) **do**
  $W \leftarrow e_i e_j^\top$
  for $e \leftarrow 1$ **to** $n + 1$ *(epochs)* **do**
    if $e > 1$ **then** $W \leftarrow M_e^\top W M_e$
    $(t_{\text{start}}, t_{\text{end}}) \leftarrow (\tau_{e-1}, \tau_e)$
    foreach *node* a *with* $W_{aa} > \varepsilon$ **do**
      for $k \leftarrow 1$ **to** K **do** $\mathbb{E}_{[u_k,v_k]}[f_{i,j}] \mathrel{+}= W_{aa}\, f(t_{\text{start}}, t_{\text{end}}, N_a, u_k, v_k)$
      $W_{aa} \leftarrow W_{aa} \exp\big(-(t_{\text{end}} - t_{\text{start}})/N_a\big)$
    end foreach
  end for
end foreach

In the present experiments the population size is shared across all nodes (`Nfixed`, a single `effective_N`); the recursion is unchanged for a per-node size vector (`Nvarying`), which simply reads N_a from the appropriate entry. The same machinery, run without the survival update and stopping at the root, yields the SNP covariance of section 2.

1.2 Adjustment for Zero Observation

It is not uncommon that one observes zero IBD segment under some circumstances. Take fig. 1 for example, if we choose to have a bin $[10cM, \infty]$ but $T_6 = 1000$ generations, the probability of having one single IBD segment of length within the bin is approximately zero. If we completely ignore the bin, then the likelihood is incomplete: we are modeling a conditional likelihood given we have some observation. Thus, when we have zero observation, we will model the probability of seeing zero IBD segment, and use a Poisson distribution to account for that.

Note that $p(L|\theta)$ corresponds to the distribution derived from sampling a position uniformly along a chromosome. Thus, this distribution is weighted by the length of IBD segments, and in order to derive the number of IBD segments we need to use the distribution derived from sampling IBD segments uniformly:

$$p(S=l|\theta) = \frac{p(L=l|\theta)}{l} \times \frac{1}{\int_0^\infty p(L=l|\theta)/ldl}$$

.

Taking the integration over length, we can obtain the expectation of length of such segments in the length range R by

$$E_R[s|\theta] = \frac{\int_u^v l \times p(l|\theta)/ldl}{\int_u^v p(l|\theta)/ldl}$$

The expected number of segments in R, by noticing that the number and length of IBD segments is approximately independent, so for a genome of length γcM, $\gamma \times \mathbb{E}_R[f|\theta] \approx \lambda_R \times \mathbb{E}[s|\theta]$:

$$\lambda_R \approx \frac{E_R[f|\theta]}{E_R[s|\theta]} \times \gamma = \int_u^v p(l|\theta)/ldl \times \gamma$$

Following a similar fashion, we could obtain the analytical form of this:

$$\int_u^v p(l|\theta)/ldl = \int_{t_1=0}^{t_2=\infty} p(g|\theta) \int_u^v (\frac{g}{50})^2 \exp(-\frac{g}{50}l)dldg$$

and

$$\begin{aligned}
\tilde{g}(t_1,t_2,N,u,v) &= s \times \int_{t_1}^{t_2} \frac{1}{N}\exp(-\frac{(g-t_1)}{N})\frac{g}{50}(\exp(-\frac{u}{50}g) - \exp(-\frac{v}{50}g))dg \\
&= s \times \exp(\frac{t_1}{N})\frac{1}{50N}\left(\left[\frac{\exp(c_u g)(c_u g - 1)}{c_u^2}\right]_{g=t_1}^{g=t_2} - \left[\frac{\exp(c_v g)(c_v g - 1)}{c_v^2}\right]_{g=t_1}^{g=t_2}\right) \\
&= s \times \tilde{f}(t_1,t_2,N,u,v) \qquad (3)
\end{aligned}$$

where s represents the survival probability till t_1, $c_u = -\left(\frac{1}{N} + \frac{u}{50}\right)$ and $c_v = -\left(\frac{1}{N} + \frac{v}{50}\right)$.

Putting the pieces together, the per-bin expected number density is accumulated by exactly the recursion of Algorithm 1 with f replaced by $\tilde{f}$, giving a per-pair, per-bin rate $\tilde{\lambda}_{i,j,k} = \sum_a W_{aa}\tilde{f}(\cdot, N_a, u_k, v_k)$. The expected segment *count* for a genome of length γ cM is $\lambda_{i,j,k} = \gamma\, n_{\text{pairs}}\, \tilde{\lambda}_{i,j,k}$, where n_{pairs} is the number of haplotype pairs compared ($\frac{n_i(n_i-1)}{2}$ when $i=j$ and $n_i n_j$ otherwise). Modeling the count as Poisson, the contribution of a bin with $\hat{f}_{i,j,k} = 0$ is the log-probability of observing no segment,

$$\log p(0 \mid \theta) = -\lambda_{i,j,k} = -\gamma\, n_{\text{pairs}}\, \tilde{\lambda}_{i,j,k},$$

which is the term added to the log-likelihood when a bin is empty (and the Gaussian term of section 1.3 is used otherwise). This keeps the likelihood proper across bins that are sometimes observed and sometimes empty, and in particular lets long-length bins inform the recent-time parameters even when they contain no segments.

1.3 Estimation from IBD Data

Now assume that we have K disjoint bins $\{b_k = [u_k, v_k]\}_{k=1}^{K}$ for IBD length, in total m populations and each population i has n_i haploid individuals. Our goal is to infer the demographic structure using the IBD data. Let $\hat{f}_{i,j,k}$ be the empirical average of fractions of genome shared through bin $b_k = [u_k, v_k]$ among $n_i \times n_j$ pairs. This is the average of fractions for population i and population j. If we have $i = j$, then the average is taken over $\frac{n_i \times (n_i - 1)}{2}$ pairs.

For a bin with a non-zero empirical fraction we model the observation as Gaussian around its expectation,

$$\hat{f}_{i,j,k} \sim \mathcal{N}\big(\mathbb{E}_{[u_k,v_k]}[f_{i,j} \mid \theta],\, \sigma^2_{i,j,k}\big)\,,$$

with $\mathbb{E}_{[u_k,v_k]}[f_{i,j} \mid \theta]$ from Algorithm 1; empty bins contribute the Poisson zero term of section 1.2. The variance $\sigma^2_{i,j,k}$ is not assumed known: it is estimated from the data by a block resampling scheme. We partition the genome into independent blocks (here 50 cM each), pool blocks across replicate genomes, and for a target genome length γ draw the corresponding number of blocks; the point estimate $\hat{f}_{i,j,k}$ is the block-pooled per-pair average.

Delete-one-haplotype jackknife. The variance $\sigma^2_{i,j,k}$ cannot be taken as the naive "variance over pairs divided by the number of pairs": the $\frac{n_i(n_i-1)}{2}$ (or $n_i n_j$) pairs are *not* independent, because every haplotype participates in many pairs. The empirical fraction is a U-statistic of degree two in the haplotypes, and the correct unit of resampling is the haplotype, not the pair. We therefore use a delete-one-haplotype jackknife. Fix a bin k and a population pair; write f_{ab} for the contribution of haplotype pair (a, b) and let $\hat{f} = \frac{1}{P}\sum f_{ab}$ be the mean over the P pairs.

Within a population ($i = j$, $P = \frac{n(n-1)}{2}$). Deleting haplotype h removes the $n - 1$ pairs that contain it, leaving $P - (n - 1)$ pairs with mean $\hat{f}_{(-h)}$. The jackknife pseudovalue and variance are

$$\theta_h \;=\; n\,\hat{f} - (n-1)\,\hat{f}_{(-h)}, \qquad \hat{\sigma}^2_{\text{jack}} \;=\; \frac{1}{n(n-1)}\sum_{h=1}^{n}\big(\theta_h - \hat{f}\big)^2,$$

where $\frac{1}{n}\sum_h \theta_h = \hat{f}$ exactly. Writing $g_h = \frac{1}{n-1}\sum_{b\neq h} f_{hb}$ for the average sharing of haplotype h with all partners, this estimator satisfies $\hat{\sigma}^2_{\text{jack}} \approx \frac{4}{n}\,\mathrm{Var}(g_h)$; the factor 4 is the variance inflation produced by the degree-two U-statistic structure and is exactly what a naive pair-level variance misses.

Between two populations ($i \neq j$, $P = n_i n_j$). The mean is now a two-sample U-statistic, so we jackknife each side independently and add the contributions. Deleting a haplotype h from population i removes the n_j pairs containing it; with the analogous quantities $\theta_h^{(i)}$ from side i and $\theta_h^{(j)}$ from side j,

$$\hat{\sigma}^2_{\text{jack}} \;=\; \underbrace{\frac{1}{n_i(n_i-1)}\sum_{h=1}^{n_i}\big(\theta_h^{(i)} - \hat{f}\big)^2}_{\text{pop } i \text{ side}} \;+\; \underbrace{\frac{1}{n_j(n_j-1)}\sum_{h=1}^{n_j}\big(\theta_h^{(j)} - \hat{f}\big)^2}_{\text{pop } j \text{ side}}.$$

This per-pair, per-bin $\hat{\sigma}^2_{\text{jack}}$ is what we call $\sigma^2_{\text{jackknife}}$ in the composite likelihood of section 3; a small floor $\sigma^2 \geq \varepsilon^2$ guards against degenerate ($n \leq 2$) cells.

2 Inference via Single Nucleotide Polymorphism(SNP)

The IBD statistic is informative about *recent* coalescence (long shared segments come from recent common ancestors), but it is nearly blind to deep, ancient structure, where segments have been broken down by recombination to below the detection threshold. Allele-frequency (SNP) data carry the complementary signal: shared genetic drift accumulated in deep shared ancestry. We model the SNPs with a Gaussian (Brownian) approximation to drift, in the spirit of `TreeMix`.

Forward model of allele frequencies. Take the demographic history fig. 1 as an example. Suppose that we know the ancestral allele frequency at `anc` (i.e. at time T_5) of a SNP is x_a, then the allele frequency at the descendant populations is obtained by propagating x_a forward in time through two kinds of step. Along a branch of duration Δt in a population of size N, drift changes the frequency by a zero-mean increment whose variance, after the standard arcsine/heterozygosity scaling, is proportional to $\Delta t/N$:

$$x_{\text{child}} \;=\; x_{\text{parent}} + \eta, \qquad \mathrm{Var}(\eta) \;\approx\; \frac{\Delta t}{N}\, x(1-x).$$

At an admixture node the frequency is the linear combination of the parental frequencies, $x_{\text{admix}} = \alpha\, x_{p_1} + (1-\alpha)\, x_{p_2}$. Because drift increments on disjoint branches are independent and mean-zero, the covariance between the (standardized) frequencies of two leaves i and j is the total scaled length of the genealogical path they *share*, weighted by the probability of being co-located:

$$\Sigma_{ij} \;=\; \mathrm{Cov}(x_i, x_j) \;=\; \sum_{\text{epochs } e} \; \sum_{\text{nodes } a} W^{(\text{snp})}_{aa} \frac{\Delta t_e}{N_a},$$

where $W^{(\text{snp})}$ is propagated by exactly the matrix recursion of section 1.1 *without* the survival/coalescence update (two lineages both being in node a contributes shared drift whether or not they coalesce), and the sum runs only up to the root (there is no extension to $T_{\max}$). The admixture mixing $x_{\text{admix}} = \alpha x_{p_1} + (1-\alpha) x_{p_2}$ enters through the same admixture matrix M_e: an admixed leaf shares drift with *both* source lineages in proportion to α and $1-\alpha$, which is precisely the f-statistic signature that distinguishes admixture from a tree.

Centering and the observed statistic. Σ above is the covariance of raw frequencies relative to the unknown root frequency, which is not identifiable. As in `TreeMix` we remove it by double-centering, i.e. expressing every frequency relative to the average over populations. With row means $\bar{\Sigma}_{i\cdot}$ and grand mean $\bar{\Sigma}_{\cdot\cdot}$,

$$W^{\text{model}}_{ij} \;=\; \Sigma_{ij} - \bar{\Sigma}_{i\cdot} - \bar{\Sigma}_{j\cdot} + \bar{\Sigma}_{\cdot\cdot}.$$

The matching observed quantity is the `TreeMix` sample covariance. For each retained SNP (minor-allele frequency above a threshold; ancestral mutations older than the root removed so that only post-root variation is used) we form the deviation $d = x - \bar{x}$ of the per-population frequency from the cross-population mean and the heterozygosity $h = \bar{x}(1-\bar{x})$, and pool over SNPs with the ratio-of-sums normalization

$$\hat{W}_{ij} \;=\; \frac{\sum_{\text{SNP } a} d_i^{(a)}\, d_j^{(a)}}{\sum_{\text{SNP } a} h^{(a)}},$$

which is a heterozygosity-weighted ratio of sums, not an unweighted average of per-SNP ratios.

Standard error. We split the retained SNPs into B contiguous windows of fixed size and form the same pooled estimator within each window, $w_{ij}^{(b)} = \sum_{a\in b} d_i^{(a)} d_j^{(a)} / \sum_{a\in b} h^{(a)}$. With $\bar{w}_{ij} = \frac{1}{B}\sum_b w_{ij}^{(b)}$, the standard error of the pooled estimate is the usual standard-error-of-the-mean across windows,

$$\hat{W}^{\text{se}}_{ij} \;=\; \sqrt{\frac{1}{B(B-1)} \sum_{b=1}^{B} \left(w_{ij}^{(b)} - \bar{w}_{ij}\right)^2},$$

floored at a small ε. These per-entry errors weight the SNP likelihood below.

Finite-sample (drift) correction. With only n_i sampled haploids the observed frequency is $\hat{x}_i = x_i + \epsilon_i$, where the sampling noise has mean zero and binomial variance $\mathrm{Var}(\epsilon_i) = x_i(1-x_i)/n_i$. This noise is independent across populations, so it inflates only the diagonal of the raw product $\hat{d}_i\hat{d}_j$: in expectation

$$\mathbb{E}\big[\hat{d}_i\hat{d}_j\big] \;=\; d_i d_j \;+\; \delta_{ij}\,\frac{x_i(1-x_i)}{n_i}.$$

After the pooled heterozygosity normalization (dividing by $\sum_a h^{(a)}$ with $h = \bar{x}(1-\bar{x})$) the per-SNP bias $x_i(1-x_i)/n_i$ reduces, on average, to the constant $1/n_i$, so the raw covariance carries a diagonal bias matrix $B_{ij} = \delta_{ij}/n_i$. We subtract it; to keep the corrected matrix in the same double-centered space as $\hat{W}$ and W^{model}, we double-center the bias before subtracting,

$$\hat{W} \;\leftarrow\; \hat{W} - \tilde{B}, \qquad \tilde{B}_{ij} \;=\; B_{ij} - \bar{B}_{i\cdot} - \bar{B}_{j\cdot} + \bar{B}_{\cdot\cdot}.$$

This removes the sampling-variance pedestal that would otherwise be read as excess private drift in each leaf.

SNP likelihood. Treating the entries of the (symmetric) covariance as approximately independent Gaussian observations,

$$\hat{W}_{ij} \;\sim\; \mathcal{N}\Big(W^{\text{model}}_{ij}(\theta),\, (\kappa\, \hat{W}^{\text{se}}_{ij})^2\Big)\,,$$

where κ is a single learned overdispersion factor that absorbs the correlation between covariance entries that the block jackknife does not fully capture. With `Nfixed` the SNP model carries no separate size parameter: Σ depends on N only through the ratios $\Delta t_e/N$, so the standalone SNP model treats N as fixed (data) and infers only the event times and admixture fractions; in the combined model below N is shared with, and identified by, the IBD term.

3 The Inference Framework and Model Selection

The data given to infer the demographic structure has two parts: the IBD data and the SNP data. The two could be treated as independent, and we will use a composite likelihood with weight to connect them

$$\log p(D \mid \theta) \;=\; \underbrace{\sum_{i\le j}\sum_{k} \ell^{\text{IBD}}_{i,j,k}(\theta)}_{\text{Gaussian if } \hat{f}>0,\ \text{Poisson-zero otherwise}} \;+\; \underbrace{\sum_{i\le j} \log \mathcal{N}\Big(\hat{W}_{ij};\, W^{\text{model}}_{ij}(\theta),\, (\kappa\, \hat{W}^{\text{se}}_{ij})^2\Big)}_{\text{SNP}},$$

where $\ell^{\text{IBD}}_{i,j,k} = \log \mathcal{N}(\hat{f}_{i,j,k}; \mathbb{E}_{[u_k,v_k]}[f_{i,j} \mid \theta], \sigma^2_{\text{jackknife}})$ for observed bins and $\ell^{\text{IBD}}_{i,j,k} = -\lambda_{i,j,k}$ for empty bins (section 1.2). Both terms are themselves *composite* (the summands are not genuinely independent); the per-entry standard errors $\sigma_{\text{jackknife}}$ and $\hat{W}^{\text{se}}$, together with the learned κ, serve as the weights that keep the two data modalities on a comparable scale. The shared parameter vector is $\theta = (\{t_e\}, \{\alpha\}, N, \kappa)$ with weakly-informative priors: exponential on the inter-event durations t_e, uniform on the admixture fractions $\alpha \in [0, 1]$, a truncated normal on N, and an exponential on κ.

We use automatic differentiation variational inference as the default fitting engine, and concretely the Stan `pathfinder` algorithm.

Variational inference (VI). Exact posterior inference requires the intractable normalizer $p(D) = \int p(D \mid \theta)p(\theta)\, d\theta$. VI sidesteps it by turning inference into optimization: we pick a tractable family of distributions $\{q_\phi(\theta)\}$ (here multivariate Gaussian in an unconstrained reparameterization of θ) and choose the member closest to the posterior in Kullback–Leibler divergence, $\phi^\star =$

$\arg\min_\phi \mathrm{KL}(q_\phi \,\|\, p(\theta \mid D))$. Because the KL cannot be evaluated directly (it contains $p(D)$), one maximizes the equivalent *evidence lower bound* (ELBO),

$$\mathrm{ELBO}(\phi) \;=\; \mathbb{E}_{q_\phi}\big[\log p(D,\theta)\big] - \mathbb{E}_{q_\phi}\big[\log q_\phi(\theta)\big] \;=\; \log p(D) - \mathrm{KL}(q_\phi \,\|\, p(\theta \mid D)) \;\leq\; \log p(D),$$

so maximizing the ELBO simultaneously tightens the approximation and yields a lower bound on the log evidence (used for model selection below). The expectations are estimated by Monte Carlo with the reparameterization trick, and gradients are obtained by Stan's automatic differentiation, so the whole objective is optimized by stochastic gradient ascent (this is the ADVI recipe). VI is biased — a Gaussian q typically understates posterior variance and can miss multimodality — but it is far cheaper than MCMC, which matters because each experiment fits hundreds of model$\times$topology$\times$replicate combinations.

Pathfinder. Plain ADVI needs a learning-rate schedule and many stochastic-gradient steps. `Pathfinder` replaces that with a deterministic search: starting from a random point it runs a quasi-Newton (L-BFGS) optimization toward the posterior mode, and at *every* point along that optimization path it forms a cheap Gaussian (Laplace-type) approximation from the L-BFGS curvature estimate. It then evaluates the ELBO of each of these candidate Gaussians and keeps the best one, so it locates a good normal approximation in a handful of gradient evaluations. We run several such paths from different starts and importance-resample the pooled draws with Pareto-smoothed importance sampling (PSIS), which corrects the proposal toward the true posterior and provides a diagnostic ($\hat{k}$) for when the Gaussian approximation is too poor to trust. Pathfinder returns both the approximate posterior draws and the best-path ELBO at a small fraction of the cost of MCMC. We cross-check the variational fits against full Hamiltonian Monte Carlo (section 4) on a subset of cases.

Model selection. Each candidate topology is a separate model fitted to the same resampled data. We compare them through their (approximate) marginal evidence, using the ELBO returned by `pathfinder` as a stand-in for $\log p(D \mid \text{topology})$. For a pair of topologies we report the difference $\Delta\mathrm{ELBO} = \mathrm{ELBO}(\mathrm{T}_a) - \mathrm{ELBO}(\mathrm{T}_b)$, and across a set of candidates we form softmax "model weights"

$$w(\mathrm{T}_r) \;=\; \frac{\exp(\mathrm{ELBO}_r)}{\sum_s \exp(\mathrm{ELBO}_s)}.$$

A central question of the simulation studies below is which data modality supplies the evidence to separate topologies: IBD is expected to resolve *recent* admixture (it produces a detectable bimodality / excess of long shared segments), whereas SNP f-statistics are expected to resolve *ancient* admixture (which has left almost no long IBD). The combined model should inherit both.

4 Simulation Studies

All data are simulated with `msprime` under a known demographic history, from which we extract IBD segments (by most-recent-common-ancestor span) and SNP genotypes. Independent blocks are pooled and resampled to emulate genomes of a range of lengths, so that every reported quantity can be shown as a function of the amount of data. Unless stated otherwise the inference engine is `pathfinder` variational inference.

4.1 Uniform Effective Population Sizes

4.1.1 One Admixture Event

We first do experiment with uniform effective population size setting. This means that all populations share the same effective population size, including the root population. We start with an easy topology, shown in fig. 1, where there's is only one admixture event happening in recent time. The true population size in this case is 6000, $T_1 = 20$,$T_2 = 40$, $T_3 = 50$, $T_4 = 200$, $T_5 = 200$ and the ratio of addition to $admixPa1$ is 0.75. We simulate 50 independent blocks of genome of length 50 cM under this demographic setting, then resample the blocks without replacement for 100 times to mimic different simulations.

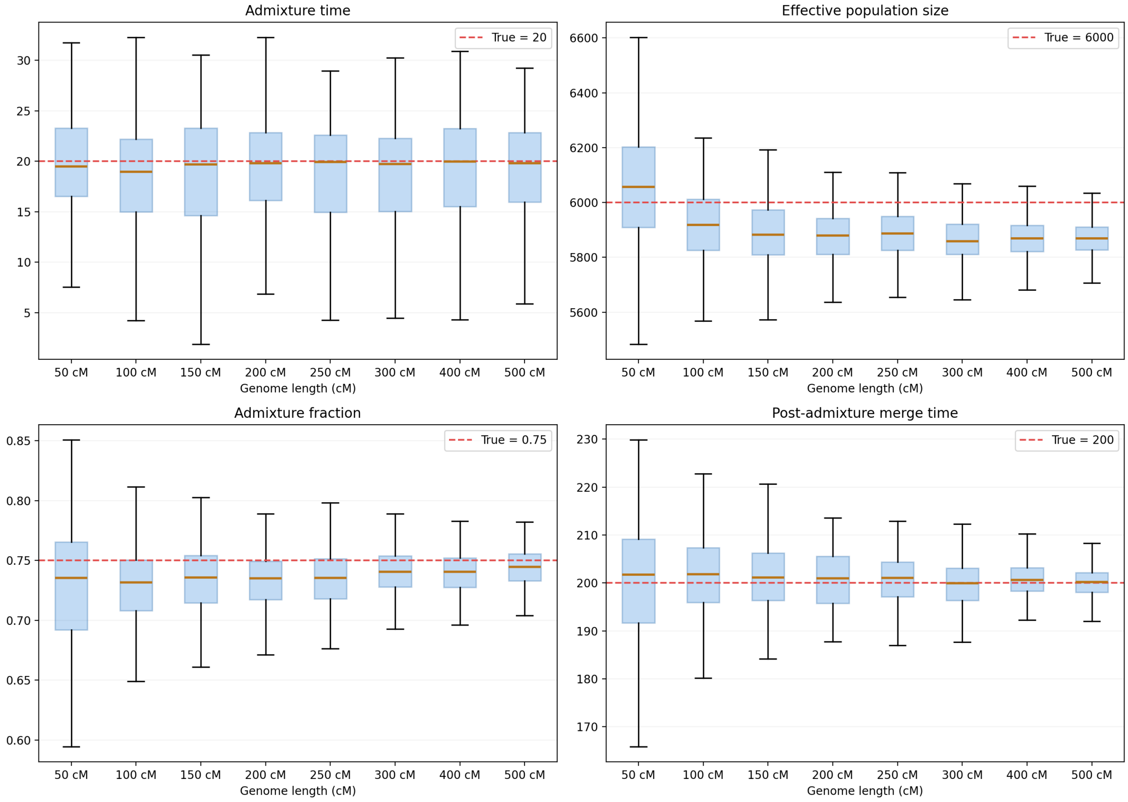

Figure 2: Posterior estimation for uniform Ne

fig. 2 shows the posterior estimates of the four free parameters (admixture time, effective population size, admixture fraction, and the post-admixture merge time) as a function of the resampled genome length. As the amount of data grows the posteriors concentrate around the truth: the admixture time and fraction are recovered first (they are driven by the abundant long IBD signal), while the effective population size and the deeper merge time need more data to stabilize. This confirms that the IBD likelihood and the zero-observation adjustment of section 1.2 together identify the recent part of the history with modest amounts of genome.

Because the variational engine is what makes the large model-selection sweeps feasible, we check that it does not distort the inference. fig. 3 compares full Hamiltonian Monte Carlo against `pathfinder` on the same resampled IBD data. The two agree closely on the point estimates; the main discrepancy is a small ($\sim$2.5%) downward bias in the effective population size under the variational approximation, which shrinks with more data and is the expected price of the Gaussian posterior approximation. Given the order-of-magnitude speed-up, we adopt the variational engine for the topology-selection experiments and treat MCMC as the reference for the parameter-recovery claims.

4.1.2 Two Admixture Events: Recent vs. Ancient

To probe whether the two data modalities carry the complementary signals predicted in sections 2 and 3, we build a four-leaf topology (a, b, c, d, all sizes uniform at $N = 10000$ haploid) that contains

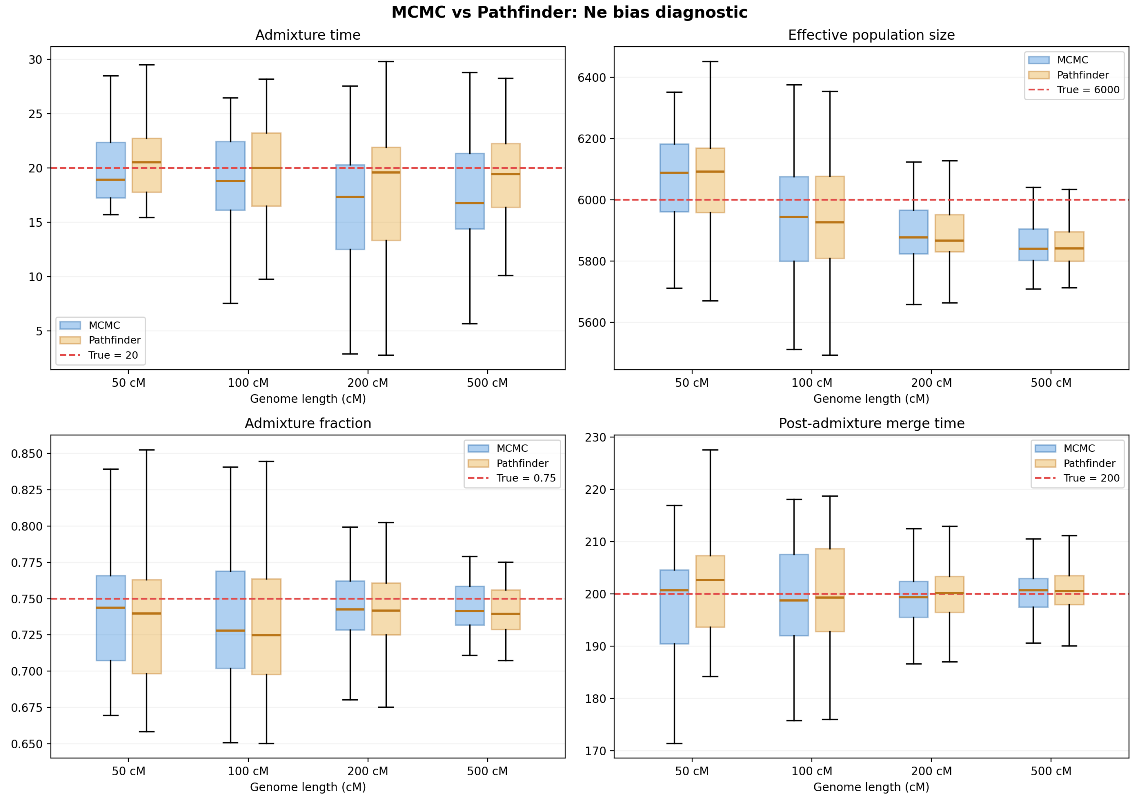

Figure 3: Bias and efficiency comparison for MCMC and Variational inference

both a *recent* admixture and an *ancient* one. The recent event is an admixture-through-b on lineage a (split at $t = 10$, merge of the donor branch into b at $t = 55$, loop closing at $t = 75$, mixing $\alpha = 0.5$); the ancient event is a 70%/30% admixture on lineage c at $t = 500$. The remaining merges occur at $t = 700, 900$ and the root at $t = 1000$.

We fit three candidate topologies (fig. 4 is the true one) to the same data: the true topology ($\mathrm{T}_{\mathrm{true}}$, both events), an alternative with the recent admixture removed ($\mathrm{T}_{\mathrm{alt1}}$, ancient only), and one with the ancient admixture removed ($\mathrm{T}_{\mathrm{alt2}}$, recent only). Each is fitted with the IBD-only, SNP-only, and Mixed models over a grid of genome lengths (50–1000 cM), with 100 resampled replicates each, and we record the ELBO and the recovered parameters. The diagnostic is the sign of $\Delta\mathrm{ELBO}(\mathrm{true} - \mathrm{alt})$: a positive value means the data favor keeping the corresponding event.

The design encodes a clear prediction. The IBD-only model should detect the *recent* admixture, $\Delta\mathrm{ELBO}(\mathrm{true} - \mathrm{alt1}) > 0$, while being essentially blind to the ancient one, $\Delta\mathrm{ELBO}(\mathrm{true} - \mathrm{alt2}) \approx 0$. The SNP-only model should show the mirror image: blind to the recent event but detecting the ancient one through the f-statistic structure of the covariance. The Mixed model should inherit both abilities.

fig. 5 confirms this division of labor. In the top row ($\Delta\mathrm{ELBO}(\mathrm{true} - \mathrm{alt1})$, the recent-admixture test) the IBD-only and Mixed models climb steadily above zero as the genome lengthens, while the SNP-only model hovers around zero — long shared segments reveal the recent loop, f-statistics do not. The bottom row ($\Delta\mathrm{ELBO}(\mathrm{true} - \mathrm{alt2})$, the ancient-admixture test) is the mirror image: SNP-only and Mixed grow strongly positive (the SNP signal is by far the largest, note the axis scale) while IBD-only stays near zero. The Mixed model is the only one positive in *both* rows, i.e. the only model that recovers the full history. fig. 6 tells the same story at the level of individual parameters: the recent split time, the two recent merge times, and the recent admixture fraction are recovered by IBD and Mixed (and missed by SNP), whereas the ancient admixture time and fraction are recovered by SNP and Mixed (and missed by IBD).

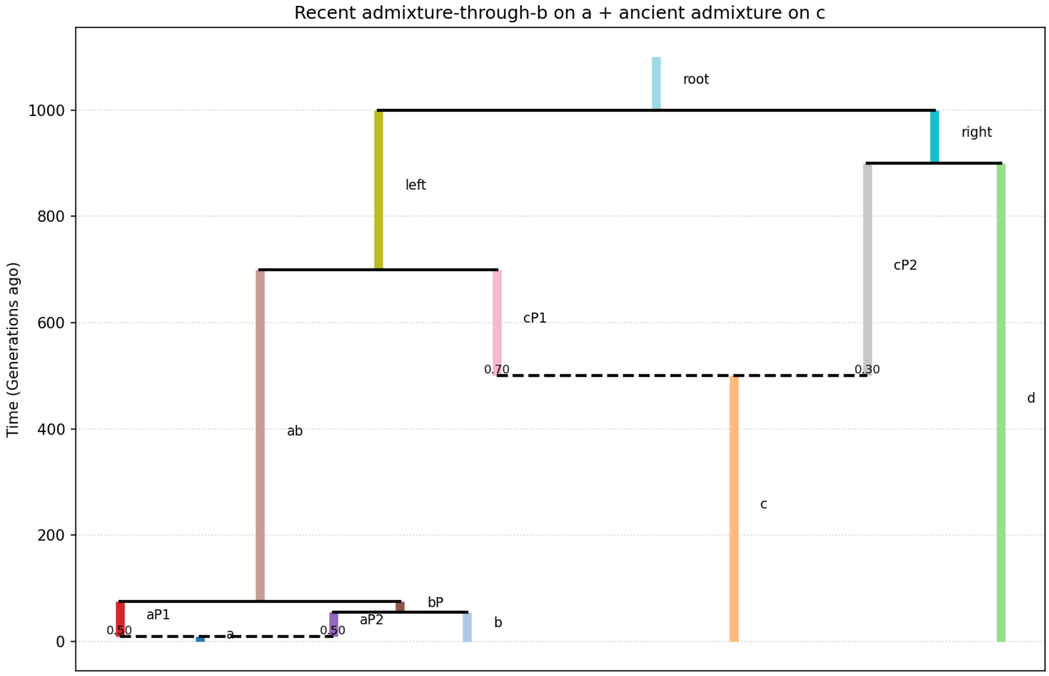

Figure 4: True topology for the two-admixture experiment: a recent admixture-through-b loop on lineage a (split at $t = 10$, donor branch merging into b at $t = 55$, loop closing at $t = 75$, $\alpha = 0.5$; dashed line = admixture) together with an ancient 70%/30% admixture on lineage c at $t = 500$. All effective sizes uniform at $N = 10000$ (haploid).

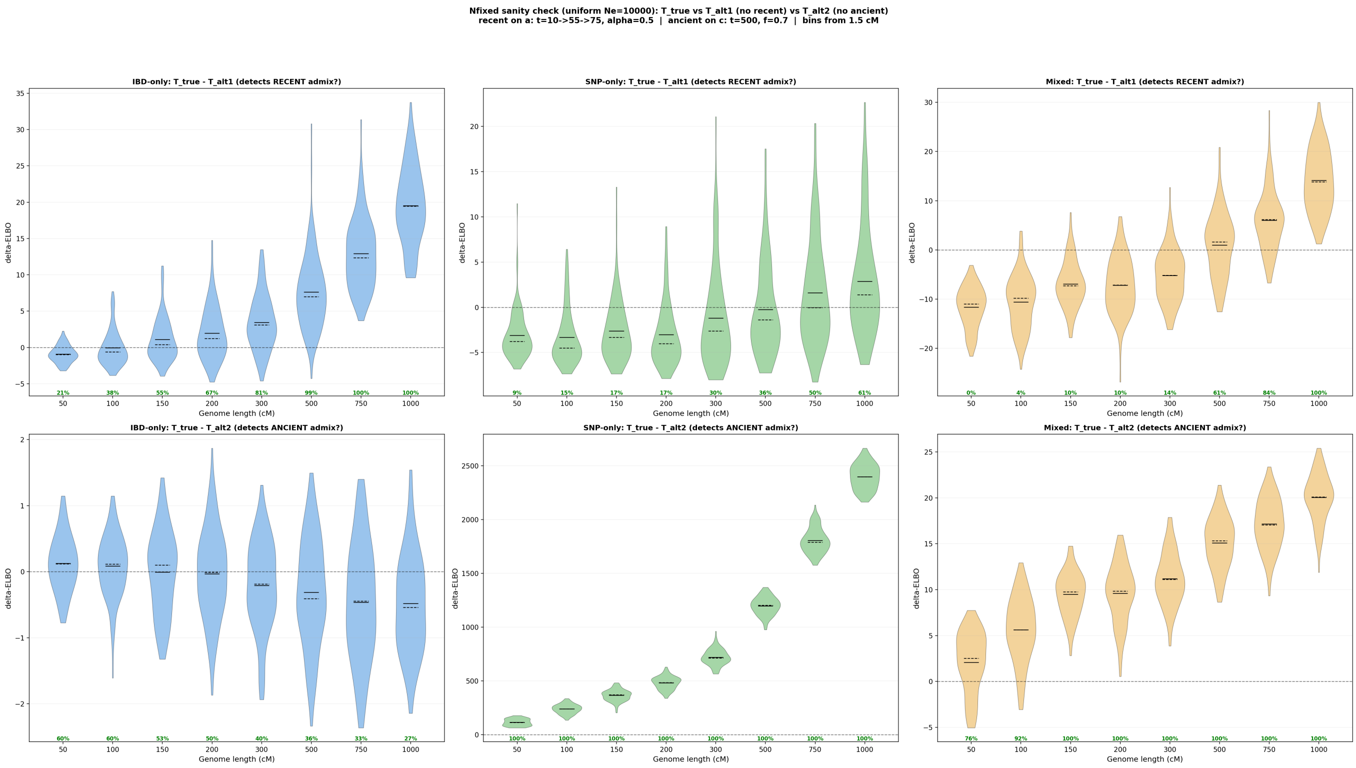

Figure 5: ΔELBO versus genome length for the two-admixture experiment, using **true (tree-sequence) IBD**. Columns: IBD-only (blue), SNP-only (green), Mixed (orange). Top row: true−alt1 (detects the recent admixture). Bottom row: true − alt2 (detects the ancient admixture). Violins are over 100 resampled replicates; the dashed line marks zero.

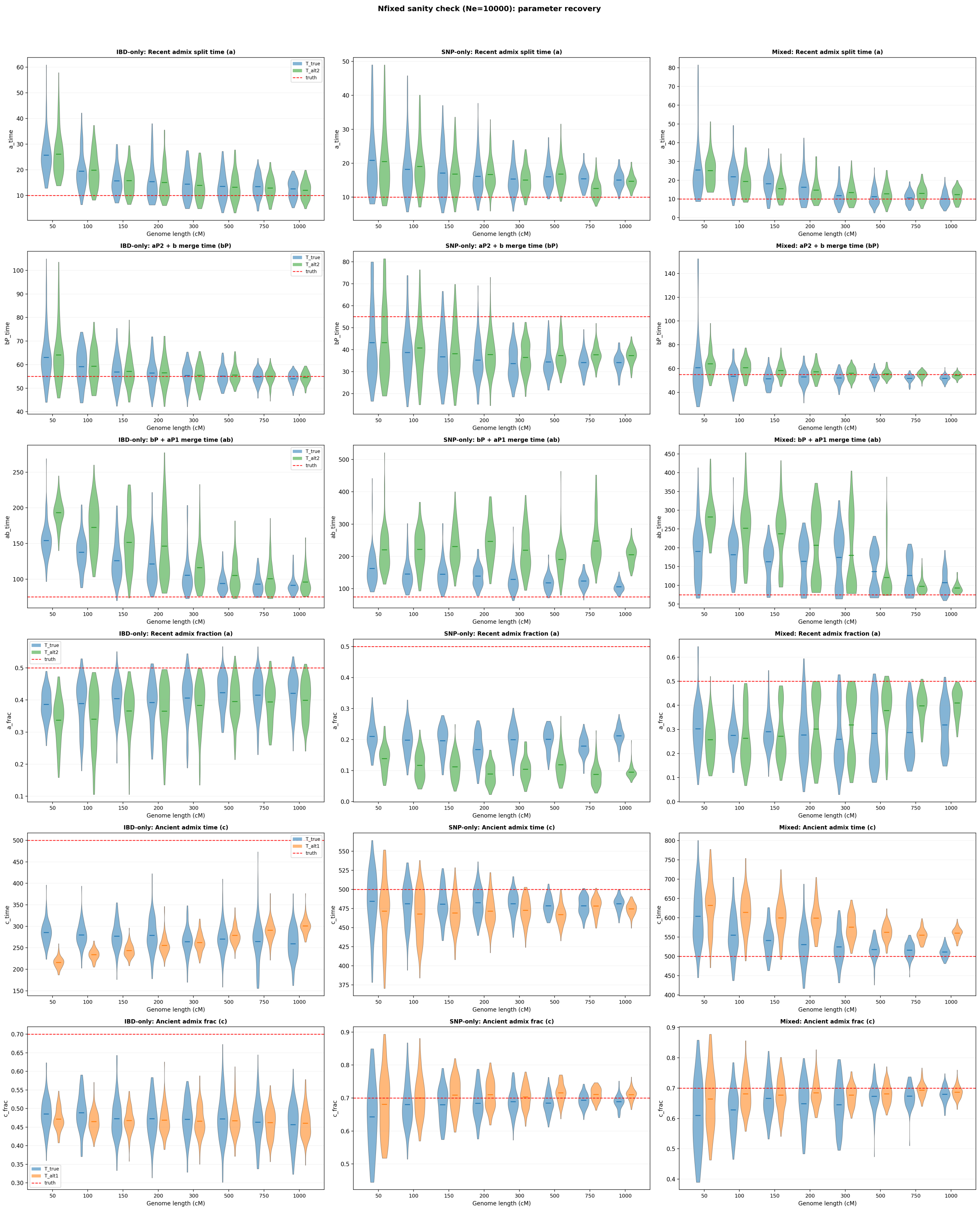

Figure 6: Parameter recovery for the two-admixture experiment (true IBD). Rows are the six event parameters; columns are the IBD-only, SNP-only and Mixed models; red lines are the simulated truth. Recent-event parameters are recovered by IBD/Mixed, ancient-event parameters by SNP/Mixed.

True IBD versus estimated IBD. The results above use the *true* IBD segments, read directly from the simulated tree sequence as spans of constant most-recent-common-ancestor. With real data the IBD segments are not observed: they must be *estimated* from genotypes by an IBD caller, which introduces phasing errors, a detection-length threshold, and boundary noise that the tree-sequence ground truth does not have. To check that our conclusions survive this, we repeat the identical experiment but replace the tree-sequence segments with segments called by `Hap-IBD` run on the simulated genotypes (VCF plus a uniform genetic map); everything downstream — binning, block pooling, jackknife variance, and inference — is unchanged. figs. 7 and 8 show that the estimated-IBD pipeline reproduces the same qualitative picture as the true-IBD pipeline: IBD/Mixed detect the recent event, SNP/Mixed detect the ancient event, and the parameter recovery degrades only modestly (slightly noisier violins and a mild attenuation at short genome lengths, where few long segments survive the caller's length threshold). This is the key robustness check: the recent-versus-ancient division of labor is a property of the data, not an artifact of having access to the exact genealogy.

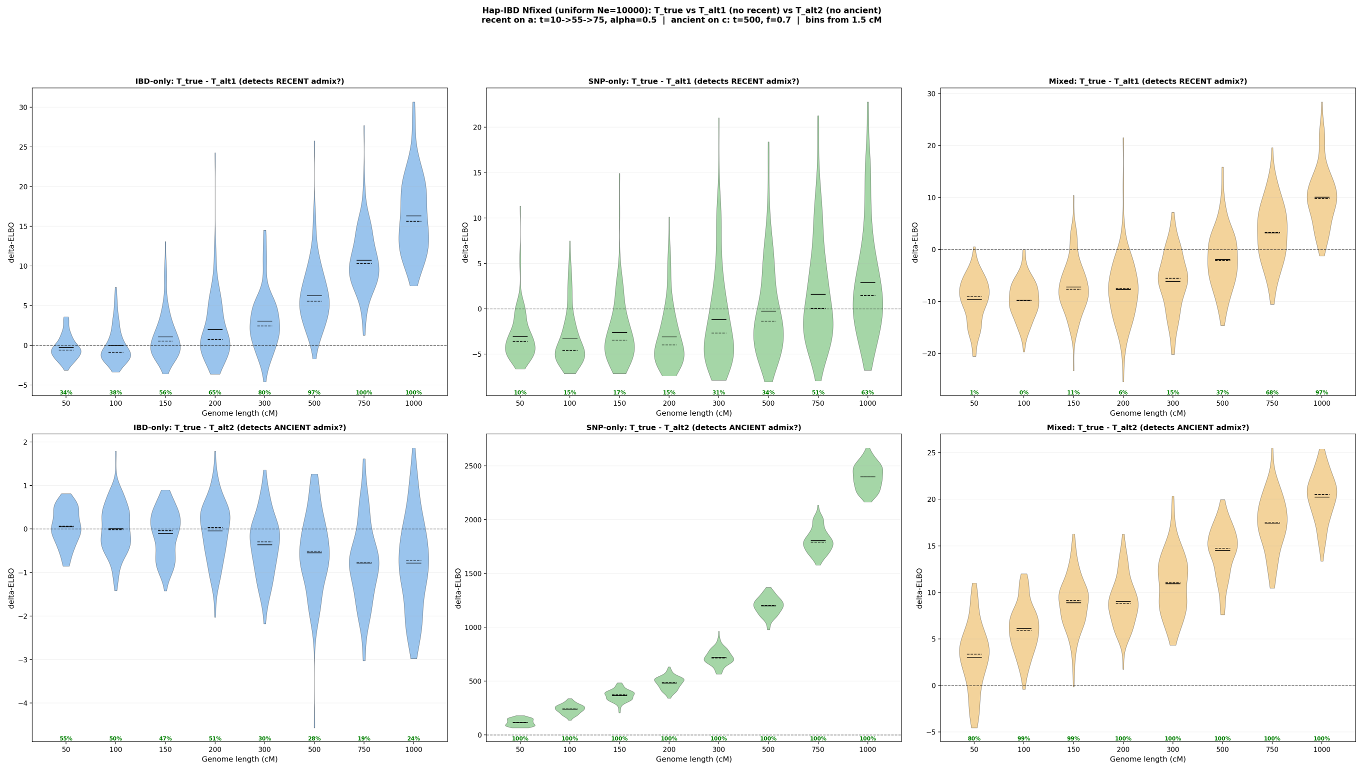

Figure 7: As fig. 5 but with IBD segments **estimated by** `Hap-IBD` from genotypes rather than read from the tree sequence. The same detection pattern survives realistic IBD calling.

4.2 Pathfinder versus MCMC (HMC)

The whole simulation programme above relies on `pathfinder` variational inference (section 3) rather than full posterior sampling, because each experiment fits hundreds of model×topology×replicate combinations. Before trusting that choice we benchmark it against the gold standard, full Hamiltonian Monte Carlo (HMC, as implemented by Stan's NUTS sampler), on the one-admixture uniform-N setup of fig. 1. We fit the same resampled IBD data with both engines over 100 replicates at each of five genome lengths; HMC is run with 4 chains, 1000 warm-up and 2000 sampling iterations per chain. The comparison has two parts: whether the two engines *agree* (bias of the posterior mean), and what that agreement *costs* (running time).

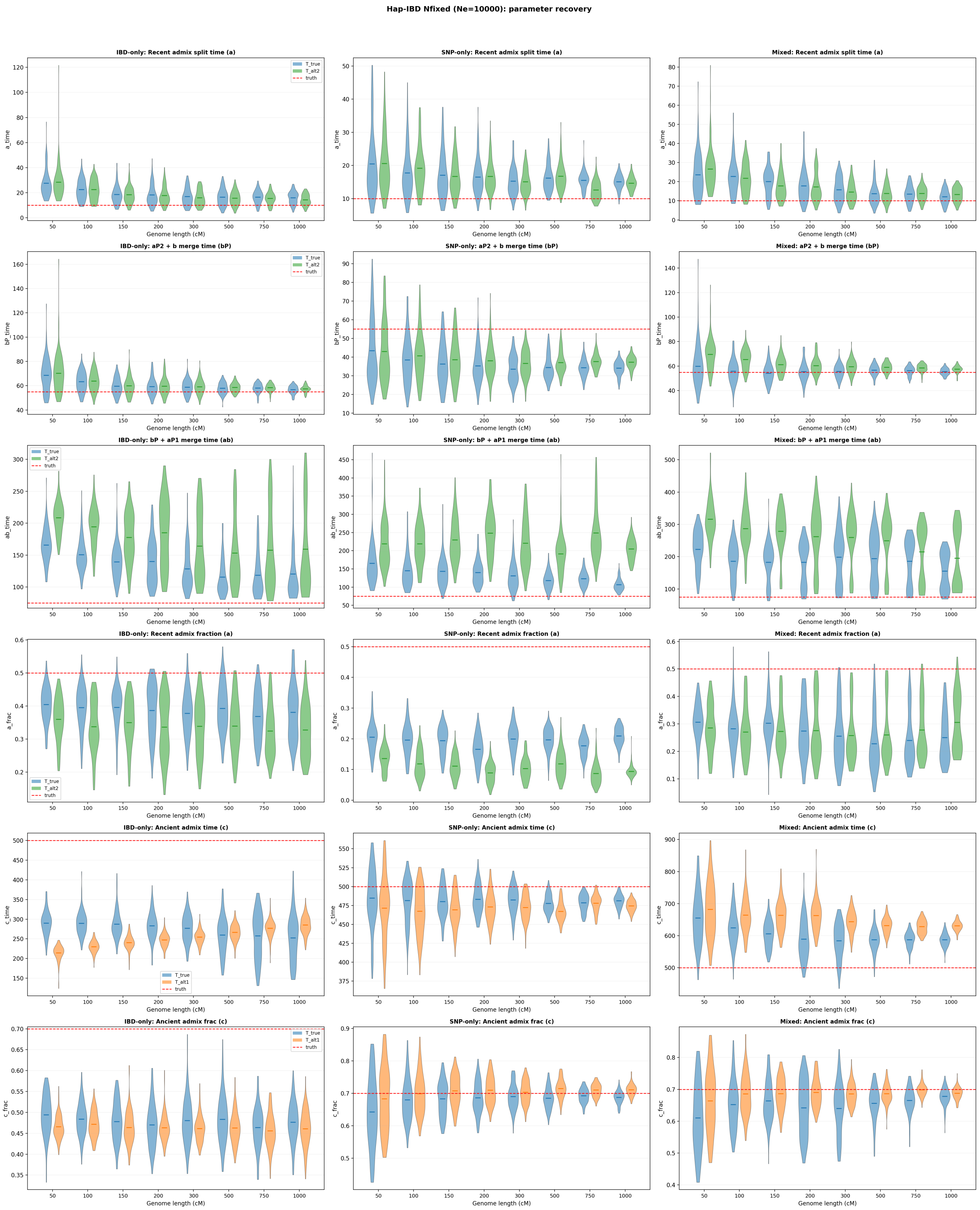

Figure 8: Parameter recovery for the two-admixture experiment with `Hap-IBD`-estimated segments. Compare fig. 6.

fig. 3 already showed the per-parameter posterior distributions side by side: the `pathfinder` and HMC boxes overlap almost entirely, so the variational approximation is recovering essentially the same posterior location and spread as the exact sampler. fig. 9 makes this quantitative and adds the cost axis. The first four panels plot the bias of the posterior mean (as a percent of the true value) for the four free parameters: the two engines track each other closely at every genome length, and the residual biases — a few-percent under-estimate of the admixture time and fraction, a $\lesssim 2\%$ wobble in the effective size and the post-admixture merge time — are *finite-data* effects shared by both methods, not artifacts introduced by the variational approximation. In other words, `pathfinder` adds negligible extra bias on top of HMC. The final panel is the running time per replicate: HMC costs $\sim$ 60–110 s and grows with the amount of data, whereas `pathfinder` stays at $\sim$ 1–2 s, a roughly 50–100× speed-up. This is what makes the large topology-selection sweeps of section 4 feasible, and it justifies using `pathfinder` throughout while reserving HMC as the reference for the parameter-recovery claims.

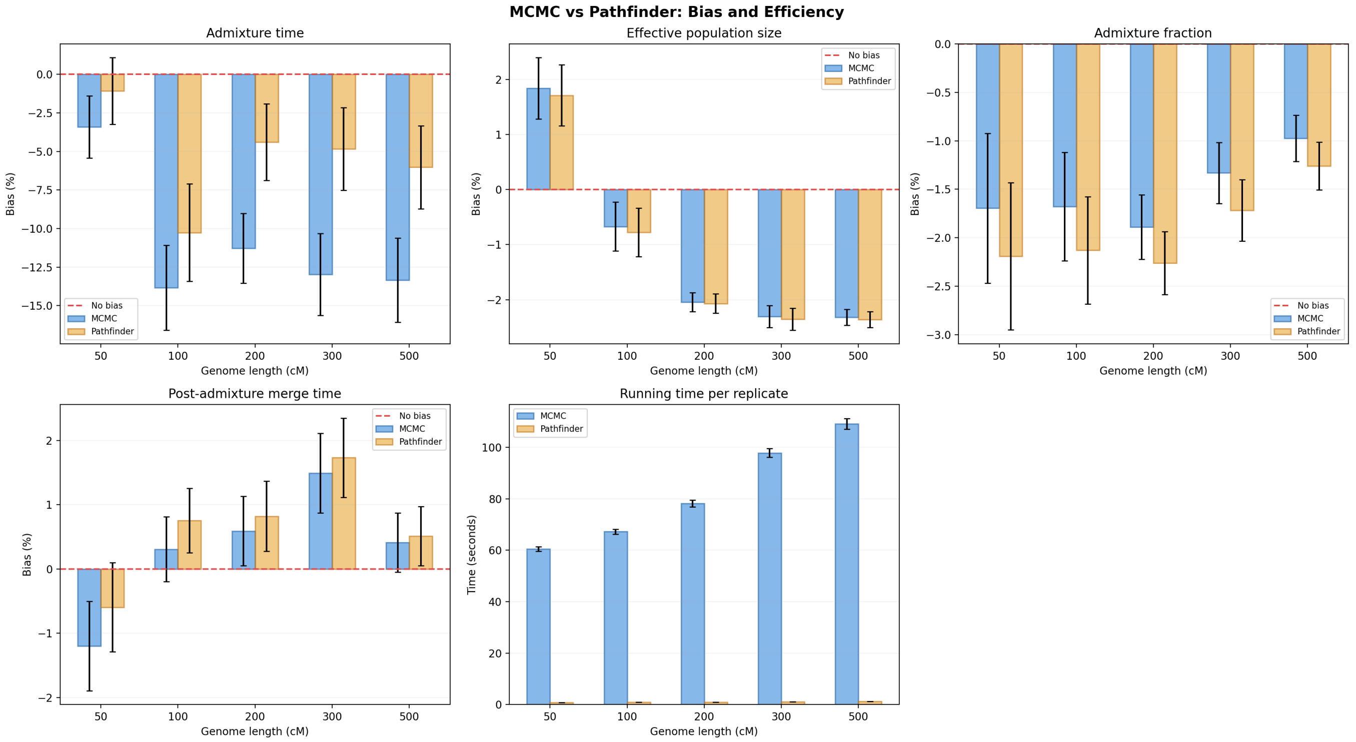

Figure 9: Pathfinder versus MCMC (HMC): bias and efficiency. First four panels: bias of the posterior mean (percent of truth) for the admixture time, effective population size, admixture fraction and post-admixture merge time, versus genome length; blue = MCMC, orange = Pathfinder, red dashed = no bias. Last panel: running time per replicate, showing the ∼50–100× speed-up of Pathfinder over HMC. 100 replicates per genome length.

4.3 Varying Effective Population Sizes

The framework extends without modification to per-node effective sizes: the recursion of Algorithm 1 reads N_a from a size vector instead of a single scalar, and the SNP covariance uses the same per-node sizes in $\Delta t_e / N_a$. The corresponding `Nvarying` models infer one size per population alongside the times and admixture fractions. The added flexibility comes at the cost of weaker identifiability: a recent branch's size and the duration of the epoch trade off against each other in the IBD likelihood, so the per-node sizes are recovered with larger uncertainty than the shared size of the uniform setting, and longer genomes are required before the deep-node sizes are pinned down.

We illustrate the varying-N regime with two five-population experiments that both carry a recent and an old admixture but differ in whether the recent event is resolvable: a *success* case where the Mixed model beats both single-data models on both events, and a *failure* case that exposes a genuine identifiability limit.

4.3.1 Success: the Mixed model beats both single-data models

The true topology (fig. 10) has a recent admixture on b at $t = 20$ (90% from the a-side, 10% from the c-side) and an old admixture on d at $t = 500$ (70%/30%), with non-uniform sizes ($N_b = N_d = 5000$, all others 15000 haploid). As before, $\mathrm{T}_{\mathrm{alt1}}$ drops the recent event and $\mathrm{T}_{\mathrm{alt2}}$ drops the old event. fig. 11 shows that here the data are rich enough that all three models eventually favor the true topology on both axes (all violins climb above zero), but the Mixed model does so *earliest and most decisively* on both the recent test (top row) and the old test (bottom row): it inherits the recent-admixture signal from IBD and the old-admixture signal from SNP, and so dominates either single-data model. This is the regime the composite likelihood is designed for — each modality supplies the event the other cannot see, and combining them strictly helps.

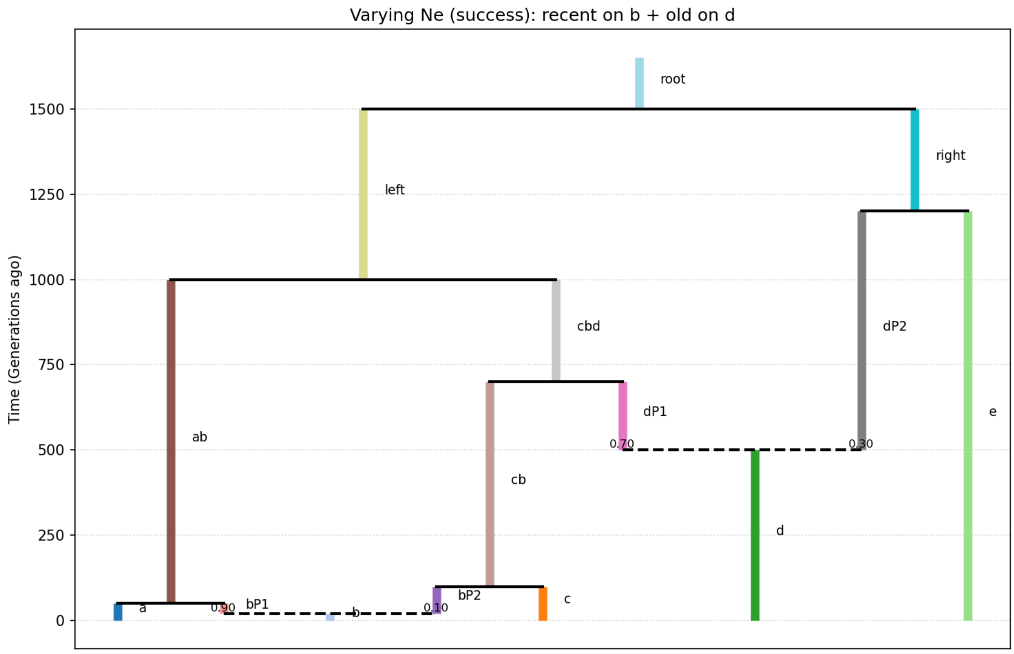

Figure 10: Varying-N success case (true topology): recent admixture on b ($t = 20$, 90%/10%) and old admixture on d ($t = 500$, 70%/30%); $N_b = N_d = 5000$, others 15000 (haploid).

4.3.2 Failure: a pulse-count identifiability limit

The second experiment is engineered to break the recent signal. Its true topology (fig. 12) replaces the single recent admixture on b with *two* pulses from the a-side (at $t = 10$ and $t = 80$) whose ancestry fractions sum to the same net 50%, and keeps an old admixture on d. The competing $\mathrm{T}_{\mathrm{alt1}}$ collapses the two pulses into a single recent pulse of 50% at an intermediate time. Because f-statistics depend only on *net* admixture proportions, not on the number of pulses (Maier *et al.*, 2023), the SNP covariance of the one-pulse and two-pulse models is essentially identical; only the IBD segment-length distribution encodes the pulse timing (a single 50% pulse caps segment lengths around 1.2 cM, leaving the 1.5–8 cM bins as a structural-zero region that only the two-pulse history can populate). fig. 13 shows the limit: on the pulse-structure axis (top row, true − alt1) every

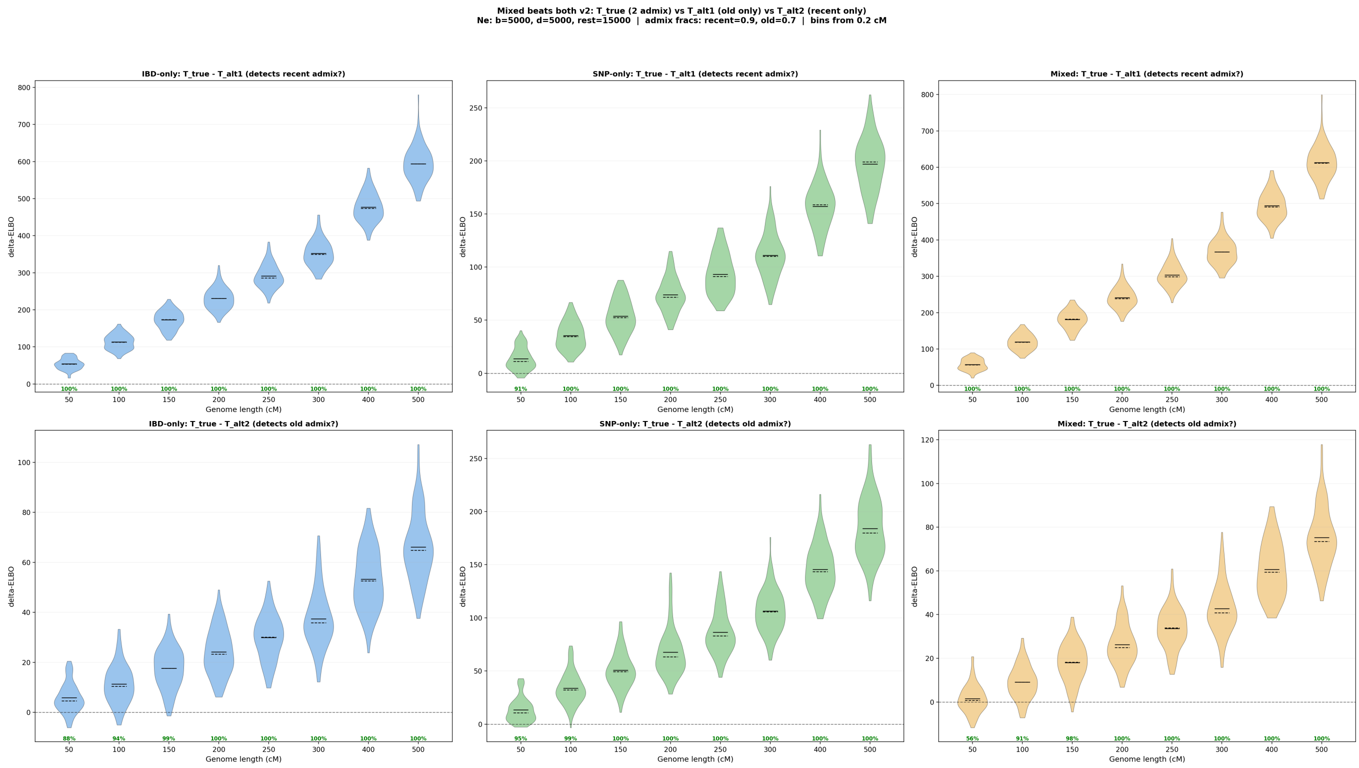

Figure 11: ΔELBO for the varying-N success case. Top row: true − alt1 (recent admixture). Bottom row: true − alt2 (old admixture). The Mixed model (orange) is positive earliest on both axes.

model — including Mixed — has ΔELBO centred *below* zero, i.e. the data prefer the simpler one-pulse explanation; the segment-length signal that should distinguish them is too weak at these sizes and sample numbers to overcome the penalty for the extra pulse. The old-admixture axis (bottom row, true − alt2) is still recovered (IBD and Mixed positive), confirming the failure is specific to pulse-count resolution and not a general breakdown. This case marks the current boundary of what the composite likelihood can resolve and motivates the finer-grained recent-length bins used in the `Nvarying` pipeline.

Together the two cases show that the recent-versus-ancient division of labor between IBD and SNP data carries over to the varying-N regime — the Mixed model dominates when the events are individually resolvable — while also delimiting it: when two histories are deliberately constructed to share both their f-statistics and (nearly) their IBD length spectrum, no amount of combining the two modalities recovers the distinction.

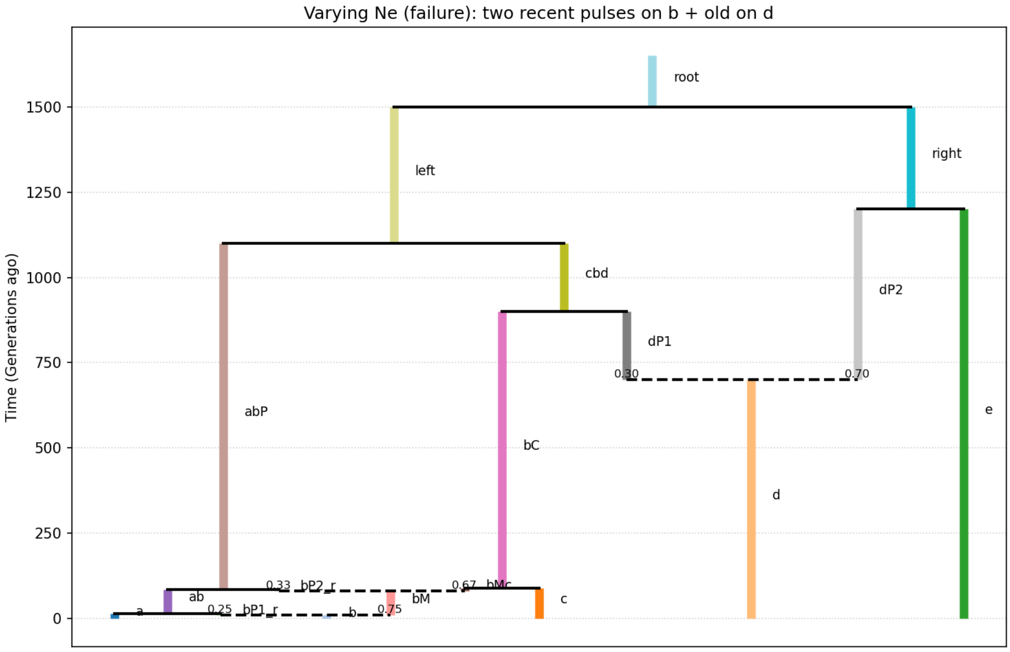

Figure 12: Varying-N failure case (true topology): *two* recent a-side pulses on b ($t = 10$ and $t = 80$, net 50%) plus an old admixture on d ($t = 700$, 30%). The competitor merges the two pulses into one, which f-statistics cannot distinguish.

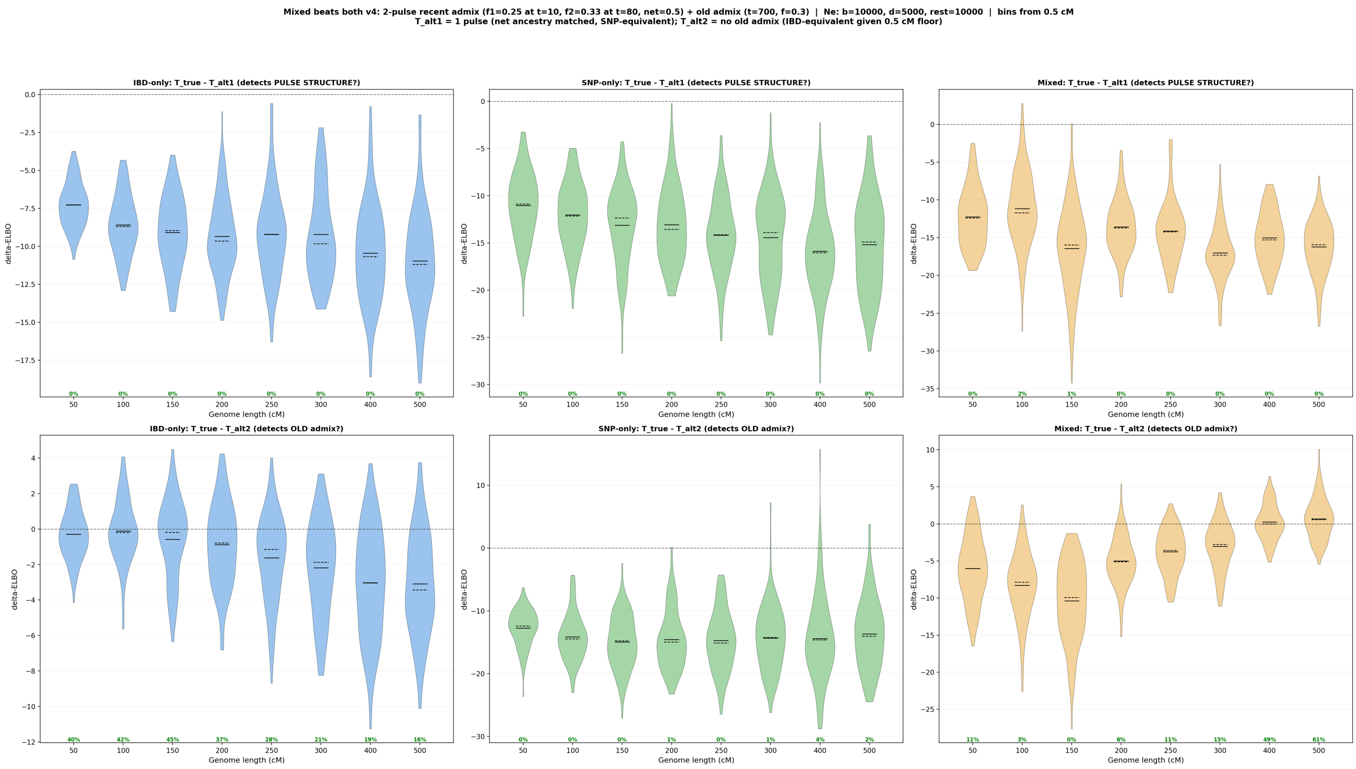

Figure 13: ΔELBO for the varying-N failure case. Top row (pulse-structure test, true − alt1): all models, including Mixed, sit below zero — the two-pulse history is not resolved. Bottom row (old-admixture test): still recovered.